Mphele Moswane

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SUMMARY

Software developer and final-year software engineering student building reliable, accessible products across web applications, backend systems and developer tools. Three-time hackathon winner with first-place builds at Geekulcha, the ITWeb Security Summit and the Mjanga AI Hackathon. Works across Java, Go, Python and TypeScript, combining systems thinking with user-centred product design and practical delivery under tight constraints. CompTIA A+ certified.

SKILLS

Languages: Java, Python, Go, TypeScript, JavaScript, Bash
Frameworks: Spring Boot, FastAPI, Node.js
Data: PostgreSQL, MySQL, SQLite, MongoDB
Engineering: REST APIs, event-driven architecture, OOP, TDD, CI/CD, Docker, Linux, Git, GitHub and ML integration

PROJECTS

Limi - Multilingual and SASL Accessibility Engine
Geekulcha Annual Hackathon 2026 | 1st Place

OCR | native speech | semantic structure recovery | offline-first learning | Demo

Transforms textbooks, photographed notes, university PDFs and videos into accessible learning formats across 12 South African languages and South African Sign Language.

- Built a universal ingestion flow for documents, photographed notes, PDFs and video content.
- Combined OCR and semantic structure recovery to preserve headings, sections and reading order.
- Delivered synchronized native speech, 3D SASL and offline-first learner modes for constrained environments.

Doria - AI Supply Chain Security Tool
ITWeb Security Summit Hackathon 2026 | 1st Place

Python | TypeScript | Node.js | XGBoost | GitHub | Demo

A zero-setup CLI that scans npm packages before installation to catch malicious, typosquatted and AI-hallucinated dependencies.

- Built a local forward proxy that intercepts installs while preserving the developer's normal terminal workflow.
- Trained an XGBoost risk model and added registry cross-checking and package hashing.

AfyaAI - Multilingual Medical Assistant
Mjanga AI Hackathon 2026 | 1st Place

Python | FastAPI | Llama 3.1 | Google Cloud TTS | GitHub

A voice-first assistant offering multilingual guidance during medical scares with severity-aware routing for urgent cases.

- Built an asynchronous backend for concurrent translation, speech and AI inference.
- Added severity parsing, guardrails, geolocation routing, language fallbacks and synchronized visemes.

Uwatu - Event-Driven Agritech IoT Platform
2026

Go | Python | FastAPI | Pydantic v2 | Nokia Network as Code API | GitHub | Demo

Models normal animal behaviour from live sensor data to surface unexplained changes that may indicate threats or early illness.

- Routed high-throughput IoT telemetry through a Go backend and built behavioural baselines from live sensor streams.
- Correlated anomalies with telecom network signals and designed for fault-tolerant operation in low-bandwidth agricultural environments.

EDUCATION

WeThinkCode_
Sep 2025 - Dec 2026

Software Development Diploma

Project-based, peer-reviewed programme covering Python fundamentals, files and configuration, REST/JSON and OAuth; Java OOP, Maven, JUnit, client-server sockets, multithreading and custom JSON protocols; and HTML, CSS and JavaScript client-side applications with asynchronous API consumption.

Built individual and team projects using git and GitHub, pair programming, agile planning, demos and technical documentation. Applied TDD, refactoring, unit, acceptance and integration testing, SQL persistence, design patterns, coupling and cohesion, CI/CD, Docker and brownfield code analysis. Selected elective: Cybersecurity.

University of Pretoria
Jan 2020 - Dec 2022

BIS Multimedia (incomplete)

Coursework combined computing, information science and visual communication, with exposure to object-oriented program design, operating systems, web technologies, information organisation and interface design.

VOLUNTEER EXPERIENCE

Peer Tutor | WeThinkCode_
Oct 2025 - Present

- Support first-year students through the WeThinkCode_ project-based curriculum by clarifying requirements, explaining programming concepts and helping them break problems into workable steps.
- Review code and guide debugging, testing and refactoring so students can identify faults independently and build stronger problem-solving habits.

CERTIFICATIONS AND AWARDS

CompTIA A+ — **Certification**
Geekulcha Annual Hackathon — **1st Place - Limi | 2026**
ITWeb Security Summit Hackathon — **1st Place - Doria | 2026**
Mjanga AI Hackathon — **1st Place Africa-wide | 2026**